Utilizing Diffusion Models to model Magnetic Field Distribution in Medium Frequency Transformers

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***Abstract*—This study investigates the efficacy of Generative AI diffusion models for modeling the magnetic field distribution in medium-frequency transformers (MFTs) faster than with the traditional finite-element-methods (FEM) approach. This study employs Diffusion Transformers (DiTs) and trains it on two-dimensional image data that is representative of magnetic field distribution in MFTs. Hyperparameters during training, architecture of DiT, and data configuration are varied to optimize for low training time, accurate modeling, and low image generation time. Results demonstrate that images with Structural Similarity Index Measure (SSIM) of 0.92 can consistently be produced by a DiT in half the time taken as FEM. Increase in training data and transformer model size in DiTs show improvements in modeling accuracy. These findings confirm that Generative AI models such as diffusion models can be considered as a viable alternative to FEM for modeling electromagnetic phenomena in MFTs.**

I. Introduction

MEDIUM Frequency Transformers (MFTs) play an integral role in a smart grid that aligns with future electricity trends. They have been shown to provide higher than 99% efficiency in power conversion [1]. Moreover, the size and weight of an MFT can be reduced by increasing its operating frequency, which is not possible to implement in conventional 50 or 60 Hz electrical transformers [1]. These attributes of MFTs set them forth as a promising renewable energy solution for the future smart grid. However, optimal design methodologies for MFTs are an ongoing area of research [2].

Modeling magnetic field distribution is an important aspect of the MFT design process. Using the magnetic field distribution in an MFT, information about eddy currents, copper losses and core losses can be deduced. This information is critical for designers when minimizing energy losses in MFT. Electromagnetic field distributions in MFTs are traditionally modeled using finite-element-methods (FEM) [3]. While FEM is highly accurate, using it to model magnetic field distribution in MFTs is computationally expensive and time-intensive. This necessitates inquiry in time-efficient alternatives to FEM in modeling the magnetic field distribution in MFTs. More time-efficient modeling would allow designers to investigate optimal MFT design methodology more effectively, and hence accelerate the journey to the future smart grid.

To offer a more practical trade-off between accuracy and time-efficiency here, this study aims to successfully harness diffusion models as an alternative to FEM for modeling the magnetic field distribution in MFTs. Diffusion models are a class of Generative Artificial Intelligence (GenAI) algorithms that noise a dataset, and train a neural network to learn the denoising procedure, so it is capable of generating high quality original samples similar to the dataset [4]. Previously, other popular GenAI algorithms like Generative Adversarial Networks have been investigated for magnetic field prediction [5], but diffusion models have not been explored within this context of modeling electromagnetic phenomena. This study addresses that gap while also exploring an alternative to FEM for modeling electromagnetic phenomena.

Moreover, the use of diffusion models in handling special data structures is an ongoing key research direction [6]. Modeling the magnetic field distribution within an MFT is a special multifaceted problem where variables are also highly interconnected; the magnetic field distribution is dependent on the number of primary and secondary windings, positions of windings relative to each other and the core, winding sizes and how close windings are to each other. By attempting to solve this complicated and specialized task, this study expands on the boundaries of capabilities of diffusion models in learning from complex datasets. Upon success, this study opens doors in investigating applications of diffusion models in increasingly complex engineering tasks; modeling heat distribution in integrated circuits for thermal management, predicting effects of varying electrical grid topologies, etc.

Thus, this study aims to contribute knowledge about the efficacy of Generative AI models in modeling electromagnetic phenomena, and expands the boundaries of diffusion model capability in learning complex tasks. This is accomplished by evaluating the hypothesis that a diffusion model with low training time, high accuracy of magnetic field modeling in MFTs, and fast image generation time can be developed. Such a diffusion model would be feasible for MFT designers to train, and would have practical advantages over FEM.

Initially, this paper formulates its modeling problem and diffusion approach briefly. Afterwards, the method section details the diffusion model, and outlines how its hyperparameters, architecture, and dataset are experimentally tuned to model magnetic field distribution of MFTs effectively. Subsequently, results of the experimentation are laid out. Then, the discussion section addresses important limitations and implications of the diffusion model approach, and ideates on future research. Finally, the conclusion section summarizes the findings of this study by focusing on the most important insights.

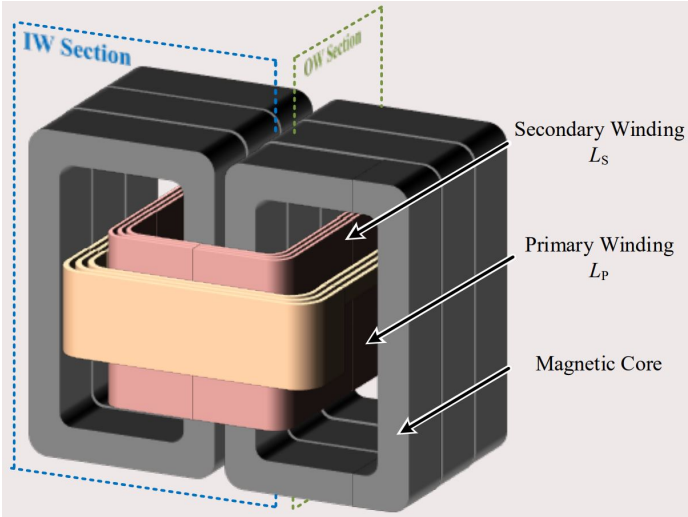

Fig. 1: Three-dimensional Configuration of a general Medium Frequency Transformer

II. Problem Definition and Approach

This section defines the modeling problem, and introduces diffusion model as an approach to solving it.

A. Two-dimensional (2D) Image Representation of Modeling Problem

Concretely defining the modeling problem is essential to solving it. Figure 1 displays the general configuration of the MFT. Within this configuration, the IW Section refers to the In-Window windings, and OW Section refers to the Out-Window windings. This study focuses only on modeling the magnetic field distribution of the two-dimensional (2D) IW-Section slice of the MFT. This is not problematic because this approach can easily be extended to modeling the OW-section of the MFT, and the MFT in three dimensions. Accordingly, the dataset is defined by two types of images. The first image type represents the MFT structure at the IW Section including number of transformer windings and their positions. The second image type represents the magnetic field distribution of the corresponding MFT structure in the first image type. The magnetic field distribution in the dataset is modeled using FEM. This representation of the modeling problem is illustrated in Figure 2.

The prepared dataset consists of 1000 pairs of images in the form shown in Figure 2. It is composed of MFT structures with varying number and positions of primary and secondary windings in the IW section. Each corresponding magnetic field distribution image in the dataset was modeled using FEM in 16 seconds on average. The goal is to utilize a diffusion model to model such an image accurately in significantly less time.

B. Diffusion Model

The working of a diffusion model is introduced in this subsection. It can be demonstrated through the forward and reverse diffusion processes in Figure 3.

In the forward process, standardized images representing magnetic field distribution of MFTs from the dataset are noised iteratively for T timesteps in the following form:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon_t \tag{1}$$

where x_t is the latent variable representation of noised image at timestep t, x_{t-1} is that at timestep prior to t, ϵ_t is an independent and identically distributed sample extracted from $\mathcal{N}(0, \mathbf{I})$ at timestep t, $\mathbf{I}$ is an identity matrix, and β_t is a constant typically linearly scheduled in the range of 0.0001 and 0.02 over the timesteps.

Thereby, the forward diffusion process is defined by the q distribution in Eq. (2), relating $\mathbf{x}$ between two timesteps:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right) \tag{2}$$

In the reverse diffusion process, a neural network $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is trained to learn the q distribution in the reverse direction, such that it can denoise pure Gaussian noise iteratively until an original sample similar to the dataset is achieved. This denoising is shown in the reverse diffusion part of Figure 3. Moreover, an image of an MFT structure can be used as a condition to guide the reverse diffusion. Eq. (3) presents the $p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)$ distribution that relates $\mathbf{x}$ between the two timesteps:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}\left(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)\mathbf{I}\right) \tag{3}$$

where $\mu_\theta(\mathbf{x}_t, t)$ and $\Sigma_\theta(\mathbf{x}_t, t)$ represent the predicted mean and variance of the Gaussian noise that will denoise the noisy latent variable $\mathbf{x}_t$ to $\mathbf{x}_{t-1}$ in the reverse diffusion process.

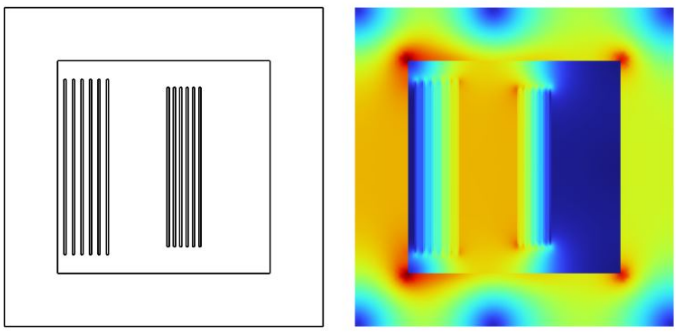

Fig. 2: Two-dimensional slice of the IW Section of a Medium Frequency Transformer, and its corresponding magnetic field distribution

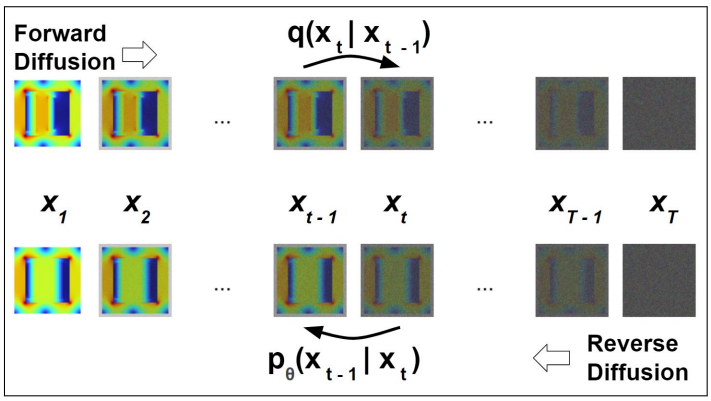

Fig. 3: Forward and Reverse processes within Diffusion Model

For the neural network to learn the features of the dataset, its weights are trained using stochastic gradient descent for minimizing a loss function evaluated between the predicted Gaussian noise and the actual Gaussian noise (based on the q distribution in the forward process). This training process requires tuning of hyperparameters that will be detailed in the Method section. The input dataset and the architecture of the trained neural network will also be varied. These aspects are critical in effectively verifying the hypothesis that a diffusion model with low training time, high accuracy, and low image generation time can be developed for the modeling problem.

III. Method

This section elucidates the experimental setup, and discusses variations in hyperparameters, model architecture, and dataset to derive insights.

A. Choice of Neural Network in Diffusion Model

The Diffusion Transformers (DiT) architecture was used for this study [7]. DiTs were chosen "for their superior visual generation capabilities, outperforming traditional diffusion models that use U-Net [8]." DiTs employ transformer, a deep learning architecture, as the backbone neural network in the diffusion process. Within a transformer, images are divided into patches, which are then embedded as tokens [9]. The transformer utilizes its multi-head self-attention mechanism to learn features across the token sequence [9], allowing for high-quality image generation. Furthermore, DiTs have scalability properties that make it easy to experiment with their architecture [7].

B. Experimental Setup

The experimental setup can be divided into two parts: hardware and software.

1) Hardware Setup

Training DiTs is computationally intensive, therefore graphical processing units (GPUs) were required. When training with images of size 256 pixels squared, NVIDIA A40 GPU was used. It has 48 GB VRAM and 48 GB RAM. When dealing with images of size 512 pixels squared, NVIDIA A100 PCIe was used. It has 80 GB RAM and 117 GB VRAM.

2) Software Setup

A comprehensive workflow for the software experimental setup is shown in Figure 4. To clarify this setup, it will be divided into six stages: i) Data Preprocessing, ii) Pre-DiT block, iii) DiT block, iv) Post-DiT block, v) Sampling, and vi) Testing.

i) Data Preprocessing: Intially, the full dataset is preprocessed at the Data Loader. Here, 1000 pairs of images are transformed by applying center-cropping to a specific size (256 or 512 pixels squared), random horizontal flipping, conversion to PyTorch tensor, and normalization of the pixel values. Moreover, the standardized images are then split and assigned to training, validation, and testing batches according to 80/10/10 split. Hence, 800 pairs of images are used for training, 100 for validation, and 100 for testing.

ii) Pre-DiT block: The standardized images representing the magnetic field distribution of MFTs are encoded into their latent representations using a pre-trained Variational Autoencoder (VAE), specifically the "stabilityai/sd-vae-ft-mse" checkpoint. This ensures that the latent representations can handle the injection of noise robustly.

At the forward diffusion block in Figure 4, the latent representations from the VAE encoder are noised till a random timestep between 1 and 1000 (T value). The Noised Latent and the corresponding timestep are inputs to the DiT. At the Patchify block, each Noised Latent is then divided into patches of fixed size 2, and then the patches are embedded as tokens. The positions of the patches are embedded using a *sin-cos* embedding. These position embeddings are also encoded into the sequence taken at Patchify in Figure 4. This token sequence is passed onto the DiT block. The timestep t is sinusoidally embedded at the Timestep Embedder.

The standardized images representing the structure of the MFT from the Data Loader were embedded using the Image Embedder block in Figure 4. Image Embedder utilized a ResNet-18 (Residual Network with 18 layers) to accurately extract high level features from the MFT structure images. These embeddings then served as context to the corresponding magnetic field distribution representation images in the

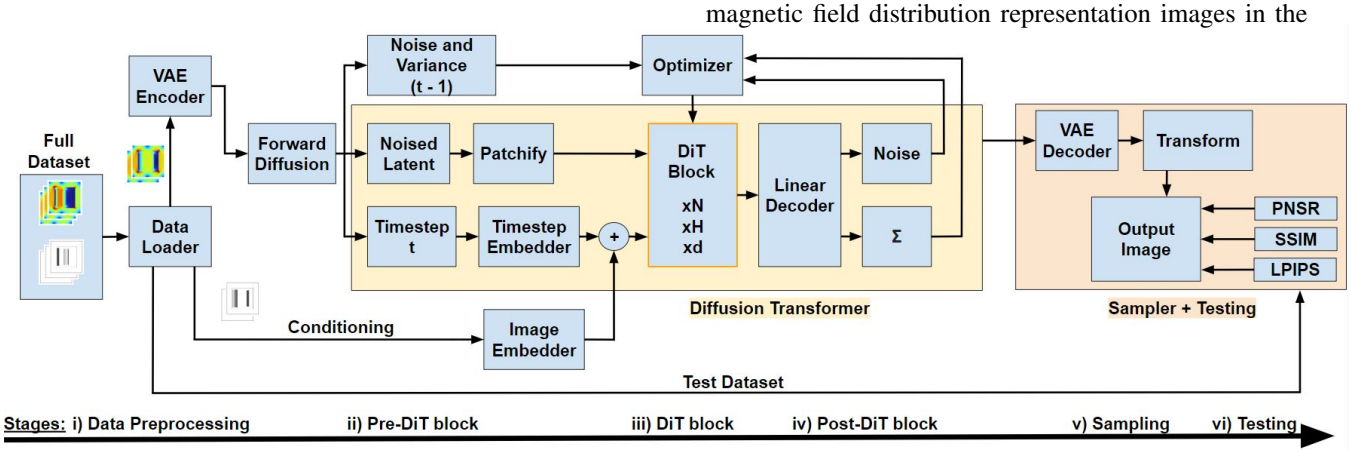

Fig. 4: Software Workflow of the Diffusion Model for Image Generation

training process of the DiT. The timestep embeddings and the context embeddings were summed and passed on to the DiT block. This acted as a condition mechanism for the DiT block.

iii) DiT block: The DiT block consisted of adaLN-Zero block for its conditioning mechanism, as it performed best for this purpose in the original paper [7]. Furthermore, a classifier-guidance mechanism was included in the forward pass of the DiT block, where both conditional and non-conditional latent representations of the noise samples were passed.

Furthermore, the DiT block had *N* number of layers, *H* number of heads, and had a dimensionality of *d*. These parameters were varied to achieve accurate modeling with low image generation time.

iv) Post-DiT block: After the token sequences passed through the DiT blocks, they were linearly decoded to extract the predicted noise mean and variance for the denoising procedure. These outputs served as inputs to the optimizer block in Figure 4. The optimizer block used a loss function to evaluate the difference between the predicted noise and the actual noise from the previous timestep in the forward diffusion block. The optimizer then used AdamW to perform stochastic gradient descent based on a chosen learning rate value. The weights in the DiT block were modified accordingly. This happened for one batch of the images. Over one epoch, training data was used to perform stochastic gradient descent for every batch. This is referred to as one training step. Afterwards, a validation step with the same process occurred with the validation dataset except that the model weights were not modified. The validation step evaluated the loss values such that they could be compared with training losses to identify over or under-fitting.

v) Sampling: After the model was trained over certain number of epochs, it could be used to sample output images. The configuration deviates slightly from the workflow in Figure 4. From the forward diffusion block, the input latent representation to the pre-trained DiT was pure Gaussian noise with timestep equal to 25, the number of sampling steps. The optimizer block was unused in this configuration. An image from the MFT structure testing dataset is passed through Data Loader and Image Embedder to condition and guide the diffusion output. After iteratively running 25 sampling steps, the final latent representation was decoded using a VAE, and transformed it from the standardized versions back to the original representation.

vi) Testing: After the DiT was trained, its ability to model the magnetic field distribution of the MFT given its structure from the testing dataset was evaluated using testing metrics. These metrics were Peak-Signal-to-Noise-Ration (PSNR), Structural Similarity Index Measure (SSIM), and Learned Perceptual Image Patch Similiarity (LPIPS). These metrics compared the ”ground truth” magnetic field distribution for the MFT structure with the image constructed through the DiT.

Sara et al. define PSNR as the ratio between the maximum possible signal power and the power of the distorting noise which affects the quality of its representation [10]. A high PSNR value reflects a higher quality of reconstructed image or higher fidelity. Equation (4) shows the PSNR calculation as follows:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}^2}{\text{MSE}} \right) \tag{4}$$

where MAX is the maximal pixel value from both images, and MSE is the mean squared error between the ”ground truth” and the diffusion model reconstructed image.

SSIM assesses reconstructed image quality by comparing the luminance, contrast and structure of the ”ground truth” and the reconstructed image [11]. PSNR measures more based on absolute error between pixels but SSIM assesses recontructed image quality on more factors relating to the structure of the images. SSIM values can range from 0 to 1, where 0 is complete dissimiliarity, and 1 is complete similarity between two images in comparison. Equation (5) shows the SSIM calculation as follows:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{5}$$

where μ_x and μ_y are the mean values of the ”ground truth” and the reconstructed images, respectively, σ_x^2 and σ_y^2 are the variances of the two images, respectively, σ_{xy} is the covariance of the two images, and C_1 and C_2 are constants to stabilize the division.

LPIPS is a metric that assess percpetual similarity between the ”ground truth” image and the reconstructed image [12]. LPIPS is different from PSNR and SSIM because it does not focus at a pixel level and uses the ”vgg” deep-neural network to extract the visual features of the images, and compute the difference between them. It is a value between 0 and 1, where 0 is perfect perceptual similarity, and 1 is complete dissimilarity.

This diffusion approach has a multitude of variables that can be investigated to achieve an optimal output. These variables can be divided into hyperparameters during training, transformer architecture, and training dataset. The approaches to how the software experimental setup will be utilized for each of them are discussed as follows. The guidelines in them are based on preliminary testing, due to the highly multifaceted nature of the modeling task.

C. Hyperparameters during Training

The initial configuration of DiT-S model based on the methods from Peebles et al. had 12 layers, 6 heads, and hidden size of 384 [7]. Image size was 256 squared pixels. Learning Rate was 1×10^{-4} with a constant learning rate scheduler. The batch size was 16. The beta scheduler was linear. The testing dataset consisted of 100 image pairs. This initial configuration was used to tune the hyperparameters. The current configuration

was changed throughout the tuning of hyperparameters. This model was trained for 15 epochs, while samples were extracted every few epochs to monitor the learning ability of the model with different hyperparameters. The hyperparameter with the best learning ability was chosen. The best learning ability here was defined as achieving a sample that resembles the MFT magnetic distribution most coherently in the earliest epochs while maintaining stable learning in the latter epochs. There was a level of subjectivity in interpreting the degree of this coherency. Stable learning was also evaluated by monitoring the training and validation losses.

1) Loss function

The loss functions find the difference between the predicted and the actual noise in the reverse diffusion process. This occurs at the optimizer block in Figure 4. With the initial configuration, three loss functions are evaluated at 15 epochs. These are: i) Mean Squared Error (MSE), ii) Mean Absolute Error (MAE), and iii) Log-cosh.

2) Beta Scheduling

After the loss function with best learning ability was evaluated and integrated to the initial model configuration, the linear and cosine squared beta schedulers were evaluated for their learning abilities. Beta Scheduling refers to the scheduling of noising throughout the timesteps at forward diffusion block in Figure 4. This was done by training the current configuration of the model for 15 epochs.

3) Batch Size

After the beta scheduling with best learning ability was evaluated and integrated to the current model configuration, batch sizes of 8, 16, 32, and 64 were evaluated for lowest time per epoch, and highest learning ability. The balance of these two at best can offer higher accuracy and lower training time.

4) Learning Rate and Learning Rate Schedulers

The learning rate finder tool from PyTorch Lightning library is used to find an initial learning rate. Then the boundaries of the learning rate at various ranges are tested for convergence and divergence through trial and error. A learning rate slightly below the divergence level was utilized, along with an exponential scheduler with gamma value of 0.97. This ensured that learning was accelerated at the start of training, but the model was allowed to converge at later stages of training.

D. Transformer Architecture

After the hyperparameters during training were configured through probing at low number of epochs, the DiT architectures with varying number of layers, heads, and hidden size in Table 1, were trained for 150 epochs and evaluated using the testing metrics and image generation time. This will provide insights to how DiT architecture affects accuracy in modeling.

E. Training Dataset

To derive effects of changes in image resolution of the training dataset, a DiT-S model was trained for 150 epochs with 512 squared pixels as well, and compared with the 256 squared pixels on testing metrics. The hyperparameters remained from the previous subsection, but were tuned slightly for a higher image resolution if necessary.

Model	Layers N	Hidden Size d	Heads
DiT-S	12	384	6
DiT-S+	12	768	6
DiT-B	12	768	12
DiT-L	24	1024	16

TABLE I: DiT models

Moreover, the effect of changes in the number of samples in the training dataset was evaluated by training the DiT-S for 150 epochs on current model configuration with 400 and 200 training samples, as opposed to 800 before. These were compared using the testing metrics.

F. Final DiT Model

Based on the results on tuning hyperparameters during training, the transformer architecture, and the training dataset, a DiT modeled was constructed to achieve highest possible accuracy in modeling the magnetic field distribution of MFTs in significantly less time as with FEM.

IV. Results

A. Hyperparameters during Training

1) Loss function

The log-cosh loss function showed more stable and faster learning than MSE and MAE as shown in Appendix A. Log-cosh also generated its first coherent sample representing magnetic field distribution of MFT at the 8th epoch, whereas the other two loss functions started showing similar behavior after 10th epoch. Therefore, it was implemented.

2) Beta Scheduling

The implemented cosine squared beta scheduling did not show stable learning despite tuning for learning rate. Whereas, linear beta scheduling demonstrated smooth and stable learning. Therefore linear beta scheduling was adopted.

3) Batch size

As shown in Figure 6 in Appendix B, when taking into account the time taken per epoch, and number of epochs, the batch size of 32 showed the best learning ability. The batch size of 8 was able to generate more coherent samples at earlier epochs but as seen in Table V it took on average 48 seconds per epoch, whereas each epoch took 17.5 seconds with batch size 32. At batch size of 64, convergence of loss function was unstable which was also reflected in the sample quality.

4) Learning Rate and Learning Rate Schedulers

A learning rate of 9×10^{-4} was applied with an exponential decay learning rate scheduler with gamma value of 0.97. It demonstrated accelerated learning at the start (relatively coherent sample at epoch 5), and smooth convergence in subsequent epochs as expected.

B. Transformer Architecture

Training the DiT architectures in Table 1 for 150 epochs derived results shown in Table II. Increasing transformer size across number of layers, heads, and hidden size show improved SSIM, PSNR and LPIPS scores. The average image

generation time dot DiT-S, DiT-S+, DiT-B, and DiT-L are 25s, 25s, 40s, and 105s respectively. An SSIM score above 0.9, PSNR above 30 dB, and LPIPS below 0.04 are achieved.

Mean Values	DiT-S	DiT-S+	DiT-B	DiT-L
SSIM	0.94	0.95	0.95	0.96
PSNR (dB)	31.94	32.20	32.37	32.67
LPIPS	0.037	0.028	0.028	0.025

TABLE II: Mean values of SSIM, PSNR and LPIPS for the models.

C. Training Dataset

Increasing image resolution from 256 squared pixels to 512 squared pixels for DiT-S model improves PSNR and LPIPS slightly, while having no impact on SSIM. This is shown in Table III. Decreasing the samples in the training dataset

Mean Values	DiT-S (256 px)	DiT-S (512 px)
SSIM	0.94	0.94
PSNR (dB)	31.94	32.39
LPIPS	0.037	0.034

TABLE III: Mean SSIM, PSNR and LPIPS for different image sizes.

produced a significant negative impact on SSIM, PSNR, and LPIPS score, as shown in Table IV.

Mean Values	DiT-S (800)	DiT-S (400)	DiT-S (200)
SSIM	0.94	0.92	0.88
PSNR (dB)	31.94	31.48	30.38
LPIPS	0.037	0.058	0.093

TABLE IV: Mean SSIM, PSNR and LPIPS for DiT-S model with training datasets of 800, 400 and 200 images.

D. Final DiT Model

Based on the results, a DiT-XS model with 6 layers, 4 heads, and 256 hidden size was trained for 150 epochs. It was able to generate images in 8 seconds (half the time as FEM) with a mean SSIM score of 0.92, PSNR of 22.52, and LPIPS of 0.050. An example of a magnetic field distribution modeled using DiT-XS model is shown in Appendix C.

V. Discussion

DiT-XS model was trained that produced SSIM of 0.92 while being able to generate samples in half the time as FEM. This confirms that Generative AI algorithms like diffusion models can model the magnetic field distribution accurately and more time-efficiently than FEM. Therefore, DiTs offer a practical trade-off between accuracy and time-efficiency. Moreover, DiT-XL achieved an SSIM score of 0.96, suggesting very accurate modeling. Increased transformer size in DiTs shows improvements in modeling accuracy. These findings can incentivise integration of diffusion models in modeling electromagnetic phenomena in MFTs as a more practical alternative to FEMs. Doing so will likely make MFT design process faster for MFT designers. This can accelerate research on MFT design methodology and fasten the path to the future electrical grid. The success of diffusion model in this application opens doors for solving increasingly complex engineering tasks.

However, DiTs with higher transformer size are less time-efficient than FEM, since the sampling time is high. This is a limiting factor. Despite the fact that DiTs can reach an SSIM of 0.96. Moreover, the conditioning of the MFT structure is not fully effective due to the details of windings in MFT structure being very small.

Areas of future research are integration of Latent Consistency Model Low Rank Adapters for reducing sampling time significantly. This would allow DiTs to be even more time-efficient with increased accuracy levels [13]. Furthermore, DiT training time and image time can also be reduced through FP4 quantization on weights and activations for DiT inference [8]. Furthermore, an investigation on training data richness and its impact on modeling accuracy of magnetic field distribution in MFTs should be explored. These future areas of research would make DiTs a stronger and more obvious alternative to FEM in modeling electromagnetic phenomena.

VI. Conclusions

MFT design methodology is an ongoing key research area which can be accelerated by making modeling electromagnetic phenomena in MFTs more time-efficient than FEM while retaining accuracy. DiTs accomplish that, while showing potential for higher accuracy with lower image generation time through future research.

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APPENDIX

APPENDIX A: LOSS FUNCTIONS

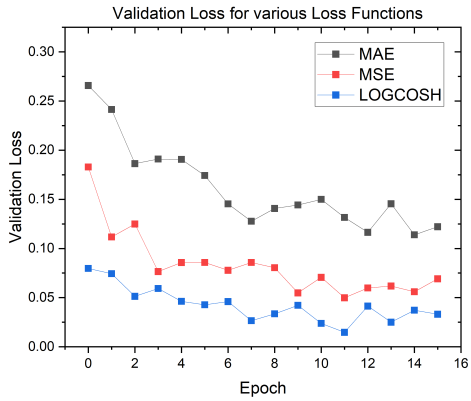

Fig. 5: Validation Loss for DiT-S model with different loss functions.

APPENDIX B: BATCH SIZE

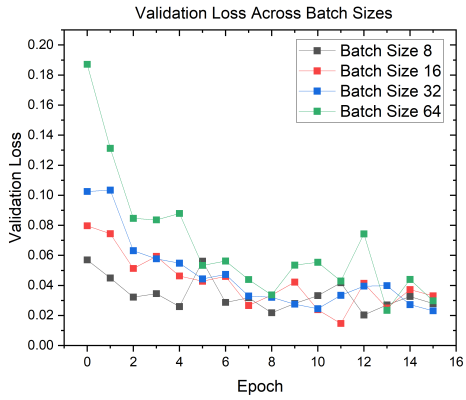

Fig. 6: Validation Loss for DiT-S model at different batch sizes.

Batch Size	Avg. completion time (s)
8	48
16	31
32	17.5
64	11

TABLE V: Average epoch completion time for DiT-S model with different batch sizes.

APPENDIX C: MODELING OF MAGNETIC FIELD DISTRIBUTION OF MFT WITH DiT-XS

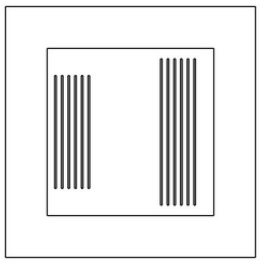

Fig. 7: MFT structure.

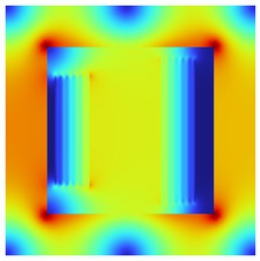

Fig. 8: Ground truth.

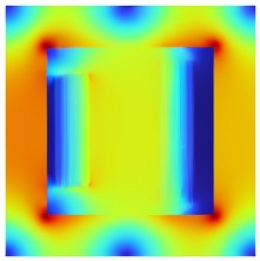

Fig. 9: Reconstructed by model.